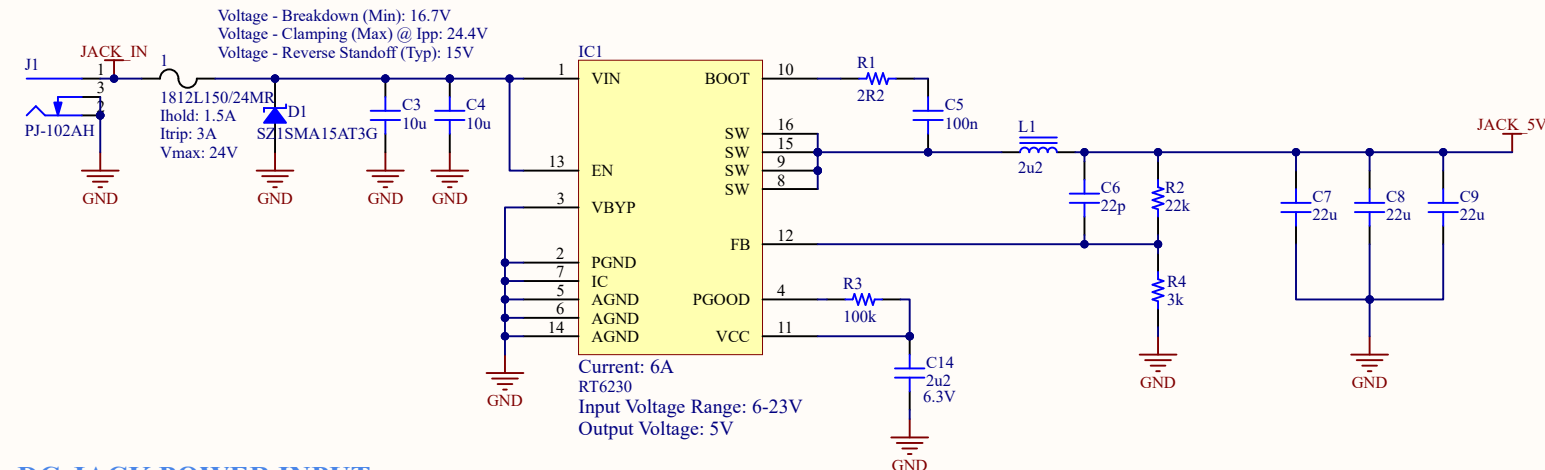


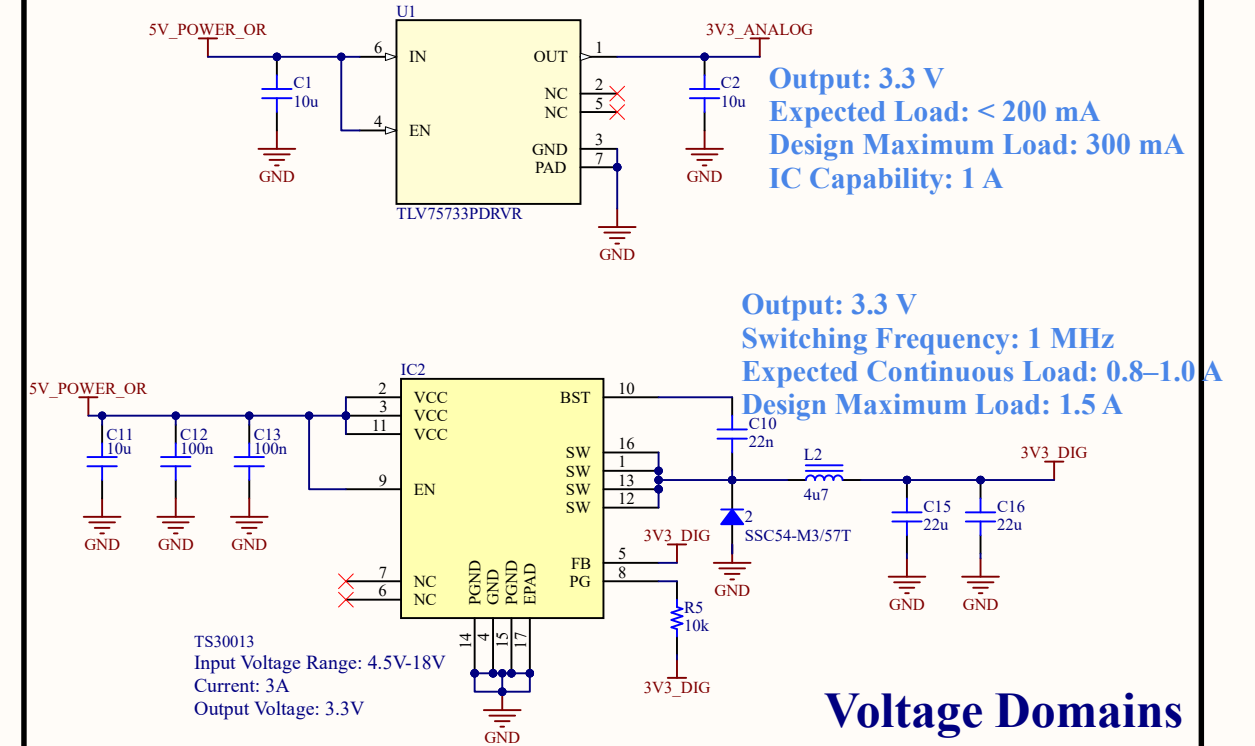
Power Section



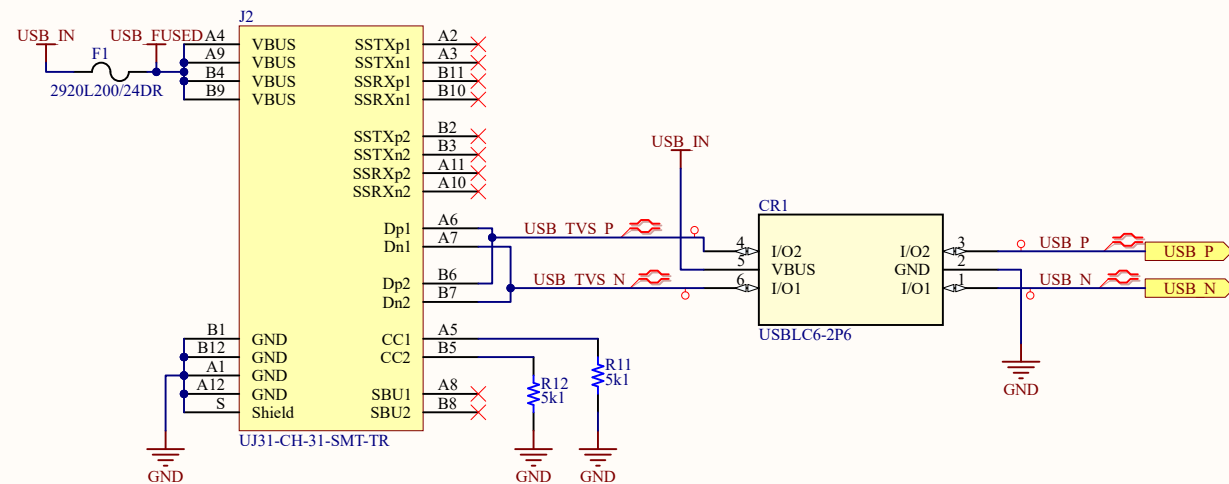
DC-JACK POWER INPUT
9–12 VDC REGULATED, CENTER-POSITIVE
Recommended Adapter: 9 V / 2 A or 12 V / 1.5 A
JACK_5V: 5.0 V / 1.5 A CONTINUOUS
2.0 A PEAK

USE CENTER-POSITIVE
ADAPTER ONLY, NO
REVERSE-POLARITY
PROTECTION !!!

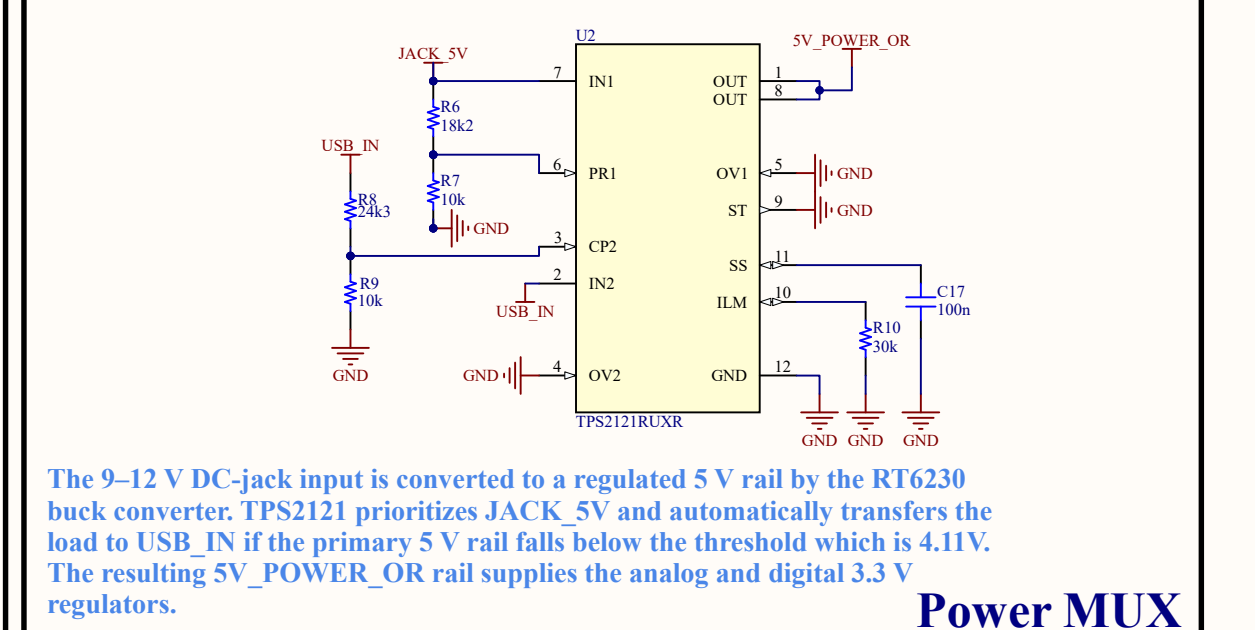
DC-Jack Input



Voltage Domains

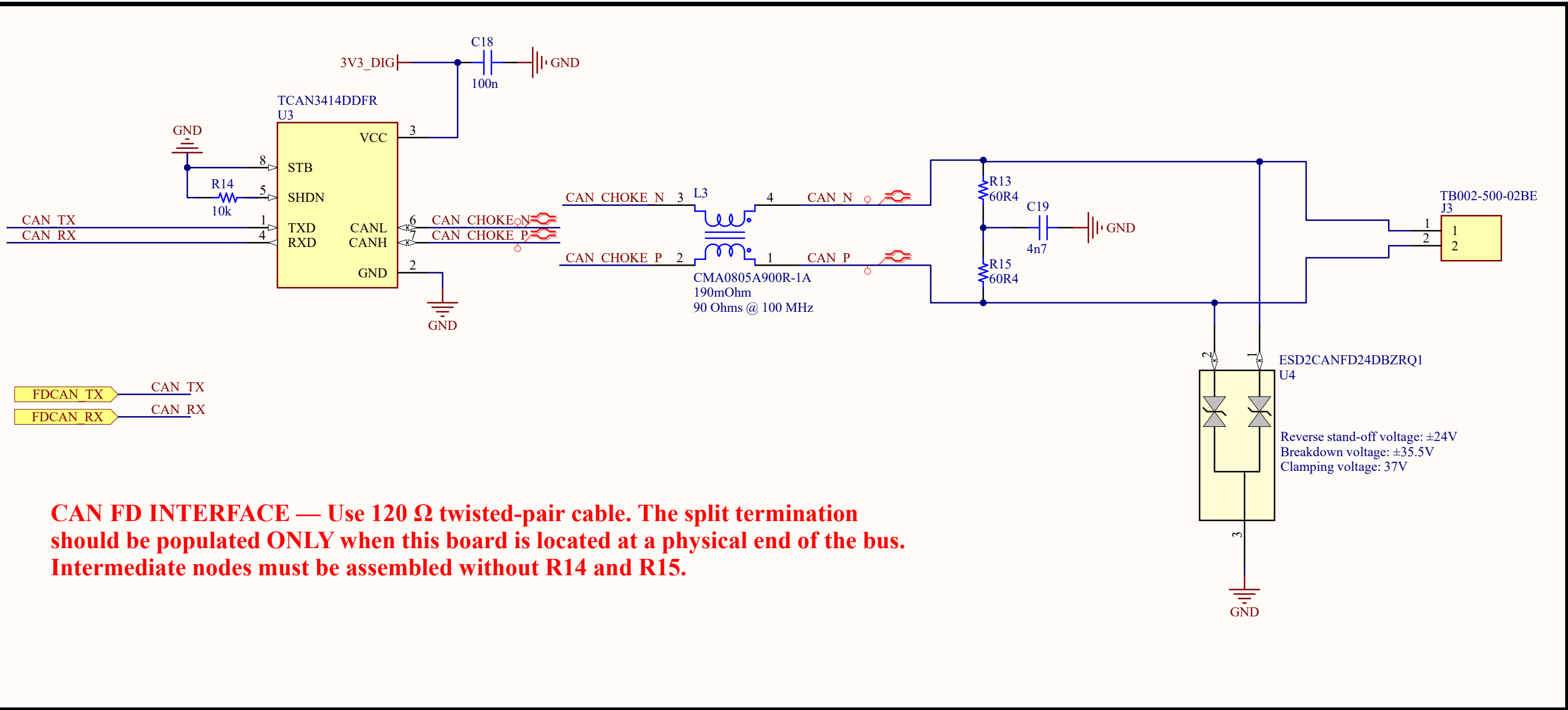


USB-TYPE-C Input



Title		
Size	Number	Revision
A3		
Date:	9.01.2026	Sheet of
File:	C:\Users\PowerNetwork\SchDoc	Drawn By:

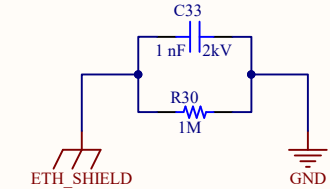
1	2	3	4	5	6
---	---	---	---	---	---



CAN FD INTERFACE — Use 120 Ω twisted-pair cable. The split termination should be populated **ONLY when this board is located at a physical end of the bus. Intermediate nodes must be assembled without R14 and R15.**

Title		
Size A4	Number	Revision
Date:	9.01.2026	Sheet of
File:	C:\Users\...\CAN-FD.SchDoc	Drawn By:

The diagram shows a 50 MHz oscillator circuit. A yellow rectangular component represents the crystal, labeled 'TZKTADSANF-50.000000 Y1'. It has four pins: pin 4 is labeled 'VDD', pin 3 is labeled 'OUT', pin 1 is labeled 'GND', and pin 2 is labeled 'GND'. A 3V3 supply is connected to the VDD pin (pin 4) through a capacitor labeled 'C20' with a value of '100n'. The output of the crystal (pin 3, labeled 'OUT') is connected to a signal line labeled 'OSCIN'. The frequency '50 MHz' is indicated below the crystal component.



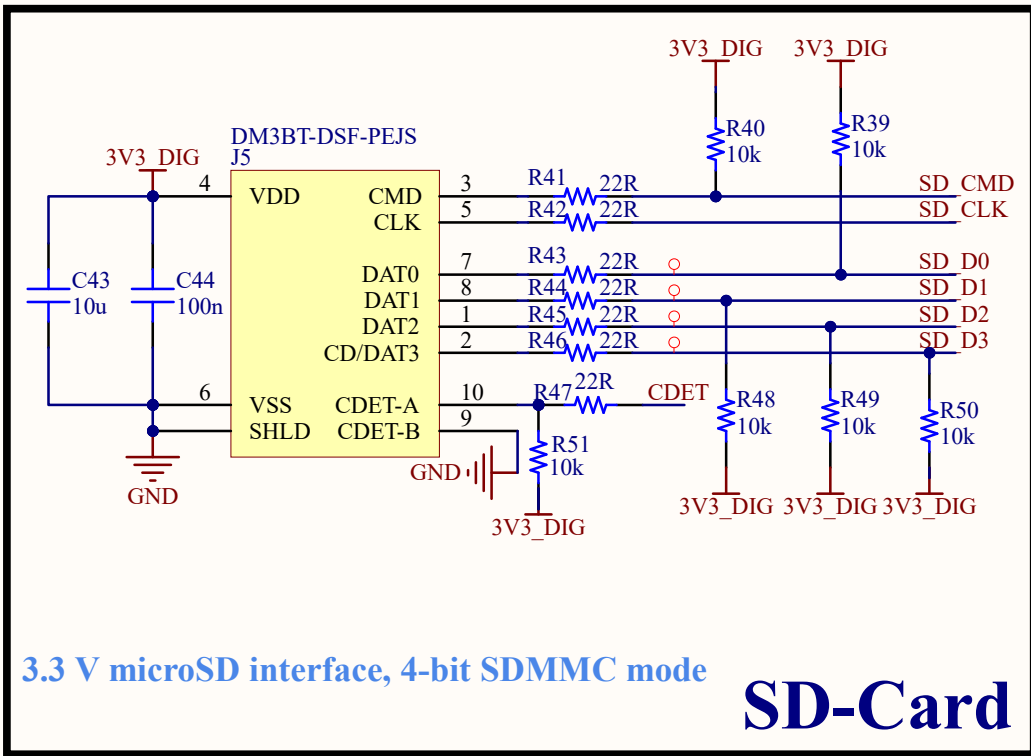
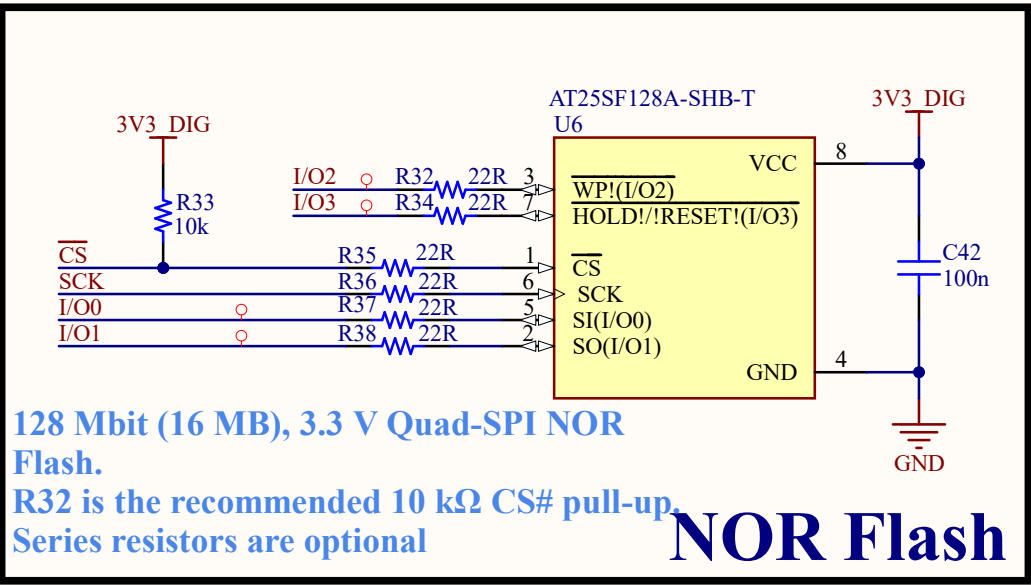
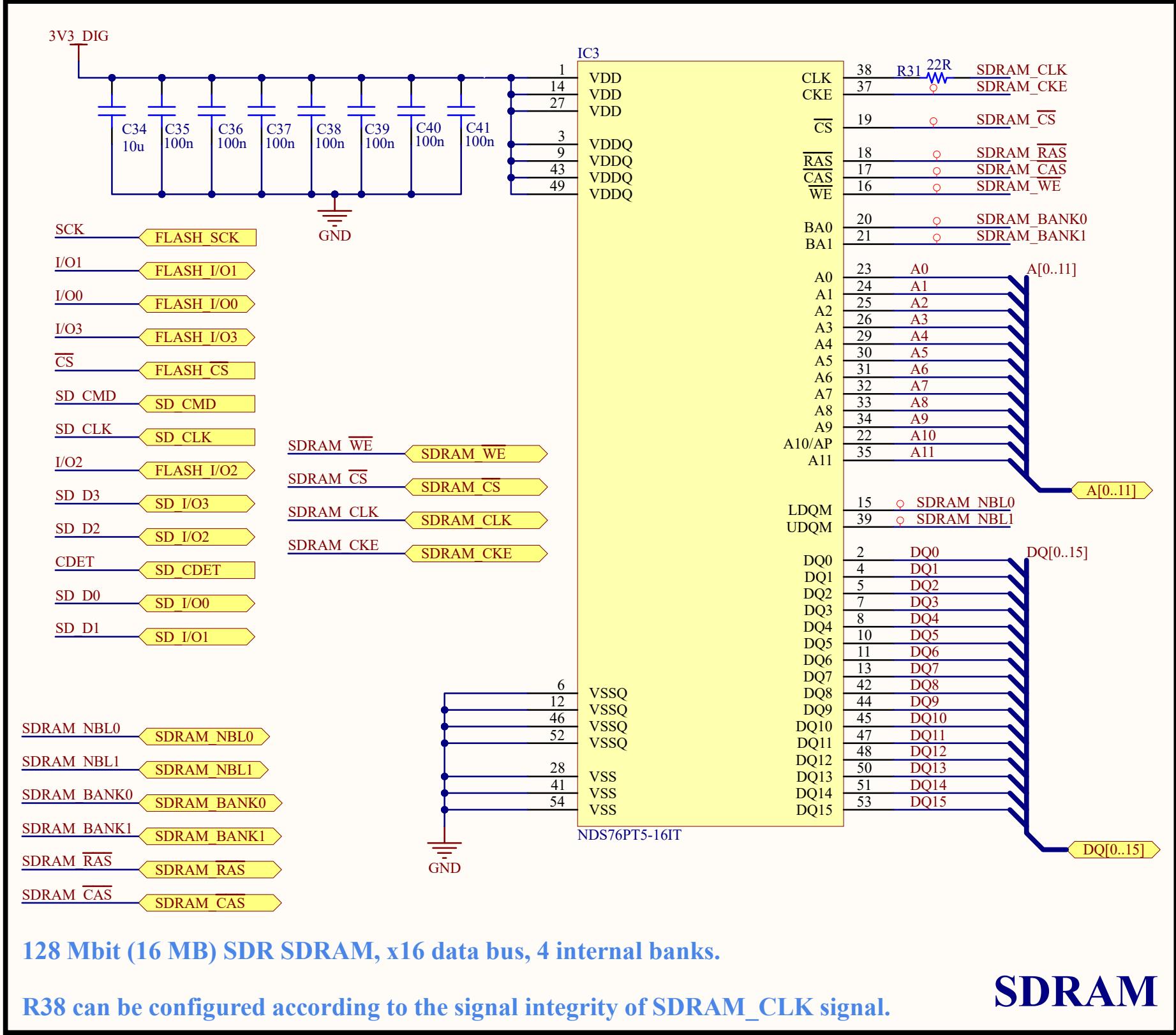
AN_EN	AN1	AN0	Forced Mode
0	0	0	10BASE-T, Half-Duplex
0	0	1	10BASE-T, Full-Duplex
0	1	0	100BASE-TX, Half-Duplex
0	1	1	100BASE-TX, Full-Duplex
AN_EN	AN1	AN0	Advertised Mode
1	0	0	10BASE-T, Half/Full-Duplex
1	0	1	100BASE-TX, Half/Full-Duplex
1	1	0	10BASE-T, Half-Duplex, 100BASE-TX, Half-Duplex
1	1	1	10BASE-T, Half/Full-Duplex, 100BASE-TX, Half/Full-Duplex

MII_MODE	SNI_MODE	MAC Interface Mode
0	X	MII Mode
1	0	RMII Mode
1	1	10 Mb SNI Mode

Ethernet PHY Circuit with MagJack

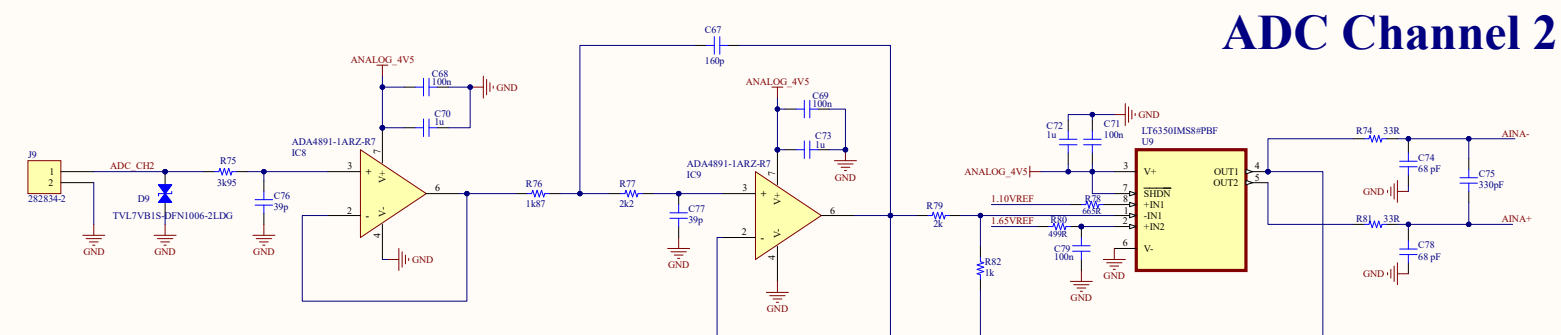
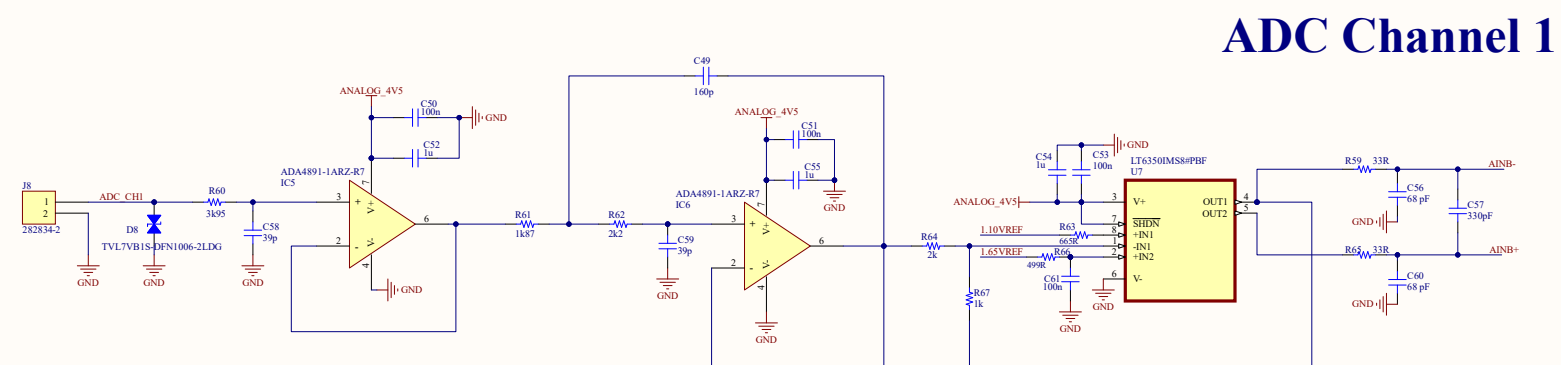
Title		
Size A3	Number	Revision
Date:	9.01.2026	Sheet of
File:	C:\Users\...\Ethernet.SchDoc	Drawn By:

Memory Section



Title		
Size	Number	Revision
A4		
Date:	9.01.2026	Sheet of
File:	C:\Users\...\Memory.SchDoc	Drawn By:

Analog Circuitry



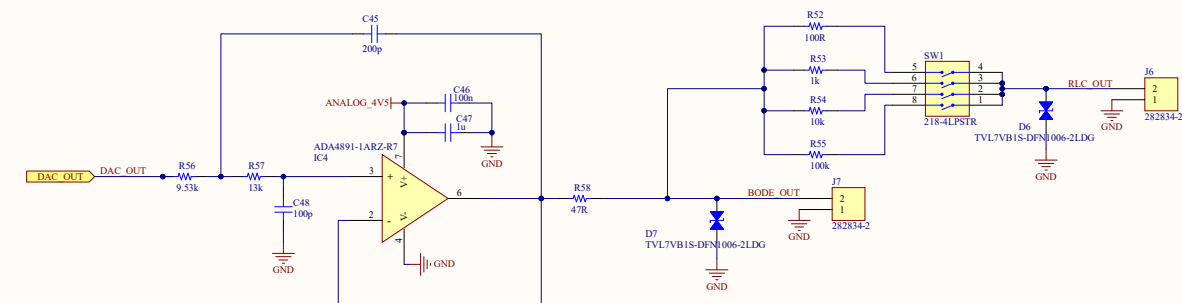
16-bit simultaneous sampling at up to 4 MSPS per channel. External input range: 0–3.3 V, single-ended. Input RC cutoff ≈ 1.03 MHz; second-order active low-pass cutoff ≈ 125 kHz. Intended precision measurement band: 10 Hz–50 kHz. ADC input network: 33 Ω / 68 pF / 330 pF.

Do not apply negative voltage or signals above 3.3 V.

ADC Inputs

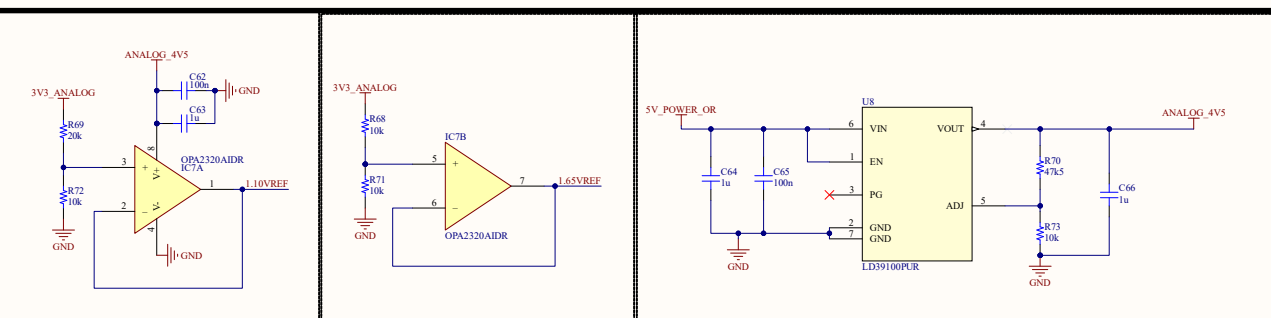
Two simultaneously sampled, 16-bit channels operate at up to 4 MSPS. Each external single-ended input is protected, buffered, low-pass filtered, level-shifted, and converted to a differential signal centered at 1.65 V before driving the AD7380. The ADC input networks use the manufacturer-recommended 33 Ω / 68 pF / 330 pF differential RC configuration. The nominal external input range is 0 V to 3.3 V referenced to board GND.

The integrated DAC signal path provides programmable excitation for frequency-response and impedance measurements. BODE_OUT is the filtered excitation output; RLC_OUT is the node after the manually selected reference resistor. Select only one reference resistor at a time. External test equipment and the device under test must share board GND. The analog interfaces are not isolated and are not permanently 50 Ω terminated.



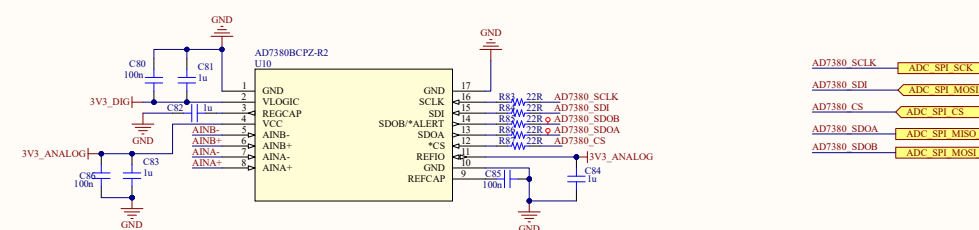
ONLY ONE REFERENCE-RESISTOR SWITCH MAY BE ON AT A TIME.

DAC Outputs



The adjustable LD39100 generates a nominal 4.5 V analog rail from 5V_POWER_OR. Buffered 1.10 V and 1.65 V bias references are derived from 3V3_ANALOG using an OPA2320. The 1.65 V node defines the differential-driver output common-mode voltage, while the 1.10 V node provides the input level-shift reference.

Voltage References



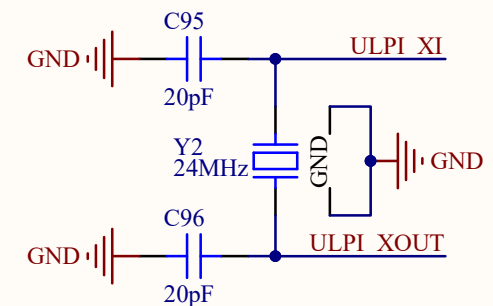
Dual simultaneous-sampling, 16-bit SAR ADC operating at up to 4 MSPS per channel. VCC is supplied from 3V3_ANALOG and VLOGIC from 3V3_DIG. REFIO is driven by the external 3.3 V analog reference rail; firmware must select external-reference operation before precision conversions.

ADC

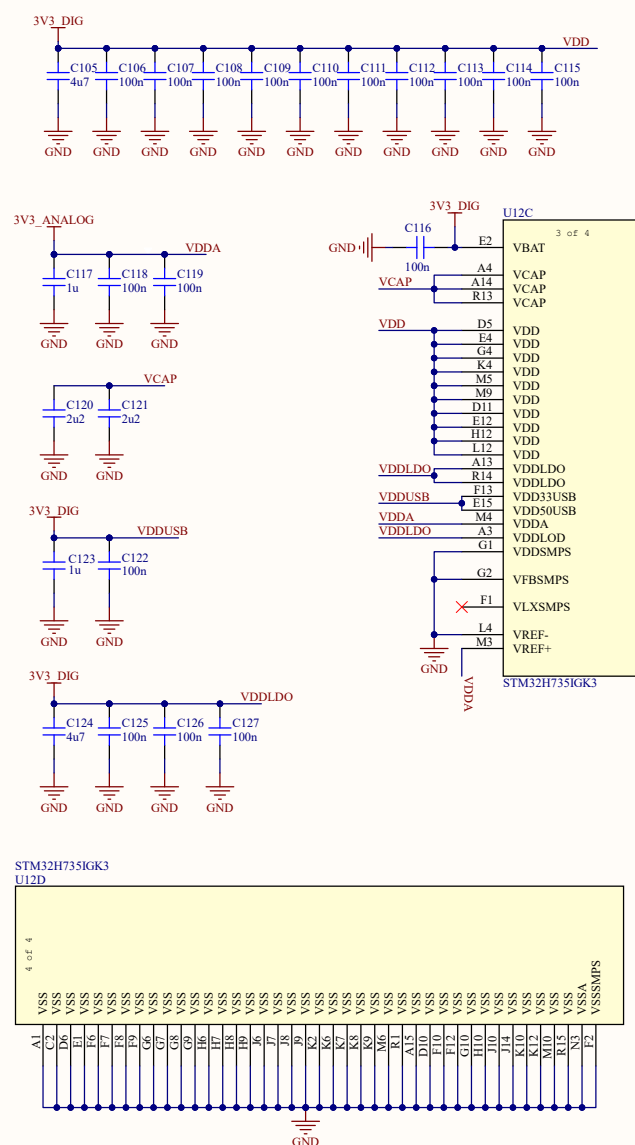
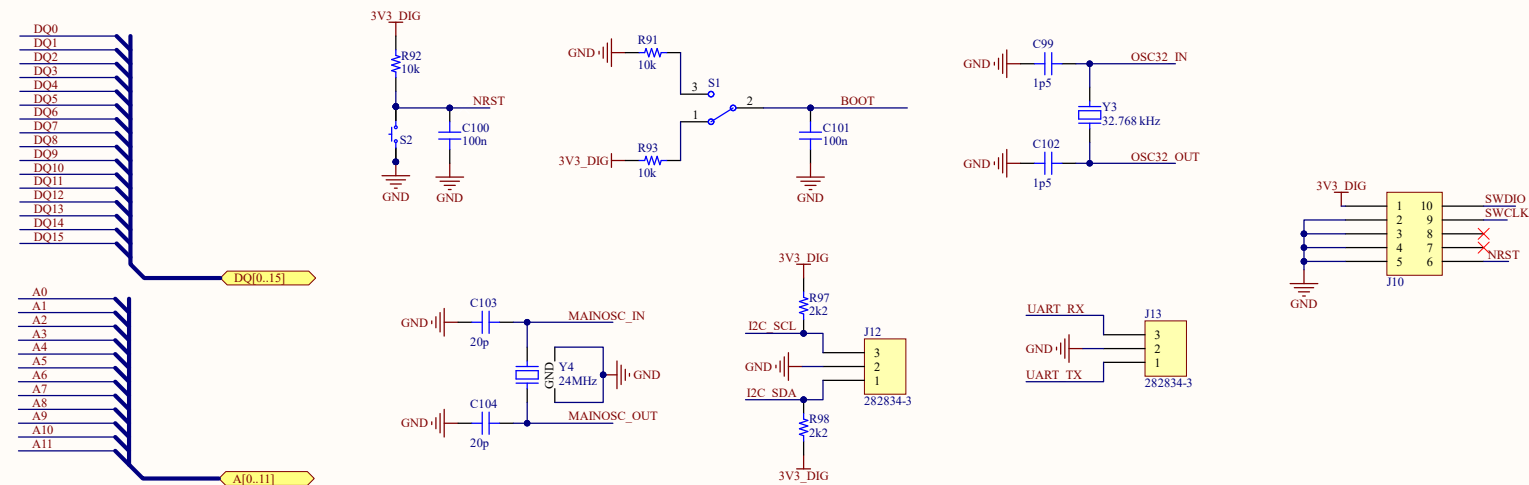
Title		
Size A2	Number	Revision
Date: 9.01.2026	Sheet of	
File: C:\Users\...\AnalogCircuitry_SchDoc	Drawn By:	

4

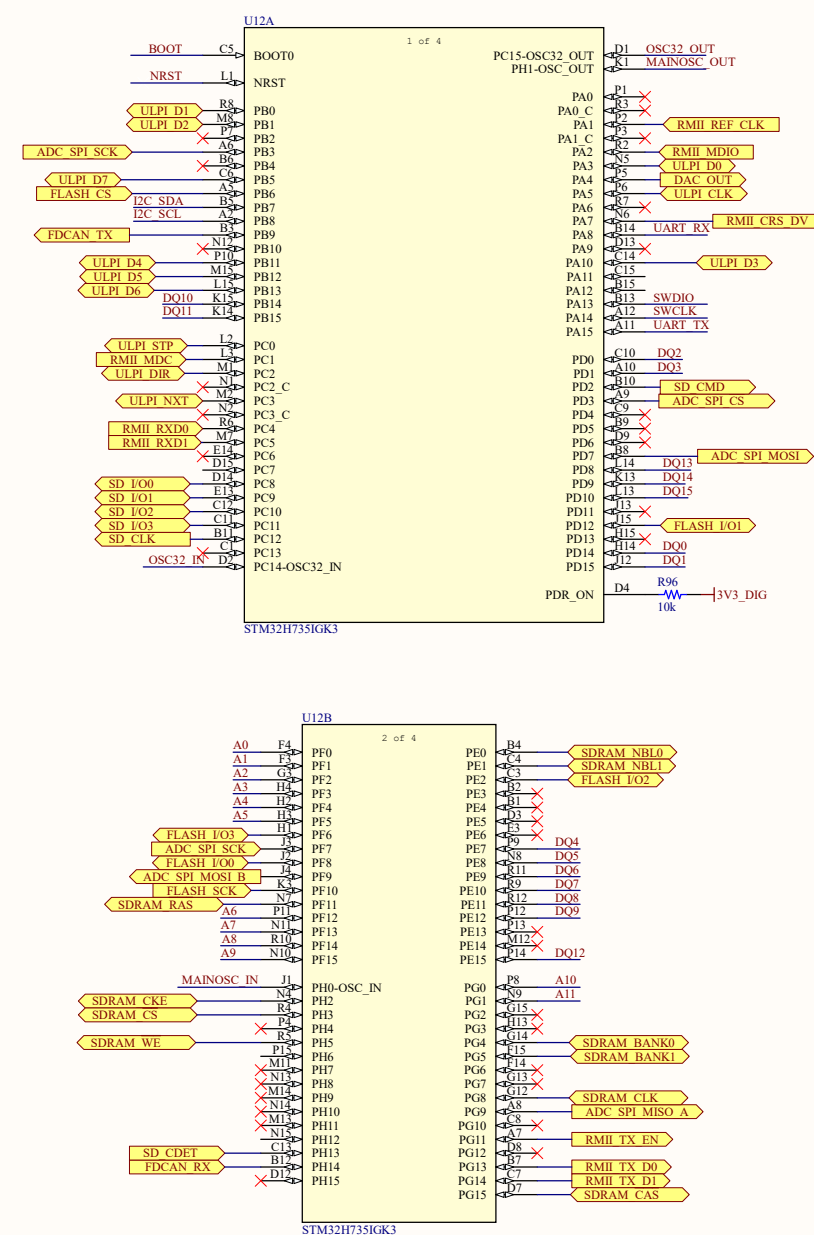
C

D

Microcontroller Section



Power Section



Title		
Size A2	Number	Revision
Date:	9.01.2026	Sheet of
File:	C:\Users\...\Microcontroller_SchDoc	Drawn By: